

9 (**short**) A car moving at speed v enters a corner with radius of curvature R centred at O. As shown in Fig. 5, the car's centre of mass is a height h above the ground, and the wheels are a width $2w$ apart. It can be assumed that R is large compared to w and h .

(a) If the car enters the corner too fast it will start to tilt. Will the wheel marked A or the wheel marked B leave the ground? [2]

(b) Show that the wheel will leave the ground if the speed of the car is greater than

$$v_t = \sqrt{\frac{gRw}{h}}.$$

[3]

(c) The car has moment of inertia $I_G = mk^2$ around a horizontal front-back oriented axis through its centre of mass. If the car actually enters the corner at $2v_t$, what is its (instantaneous) angular acceleration as it starts to tilt? [5]

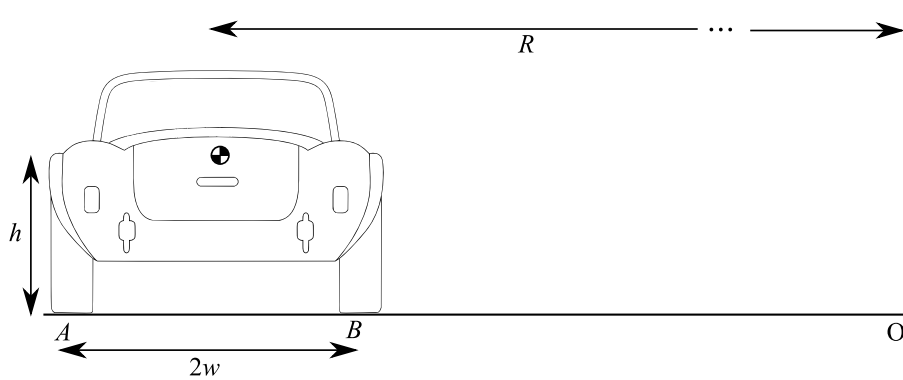


Fig. 5